

Predicting Food Delivery Time

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Abstract—This project explores the key factors influencing food delivery time using a simulated dataset from Kaggle. A combination of exploratory data analysis, interaction testing, and predictive modeling techniques—including Multiple Linear Regression, Ridge, Lasso, Generalized Linear Models, and Principal Component Analysis—were applied to identify meaningful patterns. Models were evaluated using RMSE, MAE, and adjusted R^2 . Results showed that weather conditions, traffic levels, and courier experience were among the most impactful predictors, while PCA revealed a diffuse data structure with no dominant components. The findings aim to support data-driven optimizations in food delivery operations.

I. EXECUTIVE SUMMARY

This statistical analysis seeks to investigate the predictors of food delivery time. The primary objective was to predict delivery time in minutes by creating a model using variables such as distance in kilometers, courier experience, weather, traffic level, vehicle type, and time of day.

Initial data cleaning removed missing values and all influential outliers, which ensured model reliability and prevented target leakage. Additionally, data preparation factorized and binarized all categorical variables to preserve granularity.

The key insights garnered from Exploratory Data Analysis were that distance had the strongest positive correlation with delivery time, with a correlation coefficient, r , of about 0.80. Meanwhile, weather and high traffic levels significantly increased delivery times. Most interactions were weak overall; however, some interaction was observed between weather and distance, as well as high traffic and distance.

Several different modeling techniques were used in this analysis. Multiple linear regression, Ridge and Lasso Regression, Generalized Linear Models, and Principal Component Analysis were all employed to answer the scientific question: How does a range of predictors affect the final food delivery time? The best-performing model was Lasso Regression, which showed an RMSE of 102.77.

Overall, this analysis found that delivery time is best predicted through weather and traffic variables, but particularly distance, which was found to be the strongest predictor. These findings can guide food delivery platforms when making decisions like route planning, courier assignments, and adjusting estimated delivery time to optimize customer satisfaction.

II. INTRODUCTION

This data set prompted the scientific question: How does a range of predictors affect the final food delivery

time? Found at Kaggle.com, the simulated dataset, <https://www.kaggle.com/denkuznetz/food-delivery-time-prediction>, reflects the real-world circumstances of food delivery [5].

A. Data Description

The dataset, which is simulated, was posted in January 2025. The dataset contains continuous predictors (distance, in kilometers, courier experience in years, and preparation time in minutes), discrete (order ID), as well as categorical (weather, traffic level, time of day, and vehicle type). A model fitting process will be conducted using these variables to predict the continuous target variable: delivery time, in minutes. After learning from the dataset, with our main goal for the analysis in mind, more specific questions arose about individual predictors and their influence on company efficiency. The amount of influence the predictor, vehicle type, will have on delivery time became apparent since bikes, scooters, and cars travel at vastly different rates. Can the approximate distances be covered in a similar amount of time by all vehicle types given the variability in other factors such as weather and traffic level? Another aspect that was questioned was the food delivery company's relationship with its partners. Does time of day have a strong influence on preparation time since restaurants tend to be busier during the lunch and dinner rush? One last question that emerged stemmed from the need to ensure the efficiency of the drivers: Does delivery time have a sensible correlation with distance? Results from the analysis will be critical in optimizing food delivery industry operations to identify key causes of delay in delivery, which can be used to optimize customer experience.

III. MODEL-BUILDING PROCESS

A. Data Cleaning and Preparation

The cleaning process involved identifying 117 missing cell values and removing these observations; duplicate data was checked for and not present. A suspicious observation (Order ID 446) was identified due to a car, experiencing low traffic with a short delivery distance, having an exaggerated delivery time; since this sample was also an outlier, it was removed. There appeared to be no sign of target leakage among predictors, as the available data did not offer future knowledge that would cause potential models to overestimate prediction values. The Order_ID variable, which is only used to organize

the data and trivial to our prediction-focused analysis, was removed.

Data preparation ensued once the data was rid of inadequate information. All categorical variables, with the addition of the courier experience predictor, were converted to factor variables. Furthermore, to allow for consistent measurements of p-values and coefficient estimates, each factor level was relevelled by making the level that had the most observations the base level. Binarization of the factor levels was implemented to allow for each level to be treated as a single predictor, rather than counting all levels as one predictor, which can further inquire about the difference in significance each level holds. This would prevent a circumstance where all levels are dropped from the potential model at once if one level of a categorical variable held more influence than another on the target variable; this would result in losing key information in predicting food delivery time. The cleaned data set and the newly binarized data set were combined to be used in the following steps of the model building process; the categorical variables that came from the original data set were then removed to avoid duplicate data.

B. Exploratory Data Analysis (EDA)

EDA began by looking at the distribution of the target variable, delivery time, through a histogram. As shown in Figure 1, the distribution was initially positively skewed. Although the log-transformation is ideal for positively skewed data, the square-root transformation performed better in correcting the skewness [1]. The removal of the previously mentioned suspicious outlier resulted in a less severe skew, and the addition of the square-root transformation of the target variable corrected the skew to show a relatively normal bell-shaped curve, shown in Figure 2.

Our analysis then shifted to the predictors in the data set and identifying which are influential in predicting delivery time. A correlation matrix was generated for the bivariate data exploration of the numeric variables. The heat map of the correlation matrix, shown in Figure 3, indicated a strong, positive relationship between delivery time and distance and a weak relationship between preparation time and delivery time, with correlation coefficients of 0.7959 and 0.3092, respectively; with a coefficient of -0.0275, preparation time and distance did not appear to be correlated. Preparation time was kept due to the goal of this analysis being primarily prediction; regardless of the weak correlation, being a unique predictor that is not usually taken into account for delivery time might provide information that is not captured by other variables. Bivariate data exploration for categorical variables started with viewing the mean and median values of delivery time, split by the levels of each categorical predictor. By identifying whether there is a significant difference in values between the levels of the variable, an indication of which predictors (and more specifically which levels) are most influential on the target variable can be made [2]. Snowy weather and high traffic level were both deemed the most influential in predicting delivery time since their values were significantly different (greater than

a 0.5 difference, shown by Figure 4) between their respective levels. The split box plots, represented by Figure 5, support the previously mentioned inference by showing the deviation of the high traffic and snowy weather level medians compared to other levels within their corresponding categorical variables which hold relatively similar values.

To test interactions, we chose predictors that would most likely influence each other when having an effect on the target variable [4]. The interactions that were tested were between weather, traffic level, vehicle type, preparation time, and time of day on delivery time. Figure 6 shows the decision trees that were generated for this investigation and the interactions between the following combination of predictors were seen: traffic level (high) and distance, weather (snowy) and distance, traffic level (low) and time of day (night), vehicle type (car) and weather (rainy), vehicle type (car) and weather (foggy), as well as weather (foggy) and traffic level (low); there was no interaction suggested between preparation time and distance. Although the interactions previously mentioned were all substantially weak, after further review of the interaction results and since the primary goal of this analysis is prediction, the strongest interactions will be included in the initial model, traffic level (high) and distance along with weather (snowy) and distance. This was done to prevent losing potentially influential data; model selection will navigate through insignificant interactions.

Overall, the most significant predictors indicated were distance, traffic level (high), weather (snowy), and the aforementioned interactions. The algorithms that were, subsequently, implemented are the Multiple Linear Regression, General Linear Model, Lasso Regression, Ridge Regression, and Principal Component Analysis.

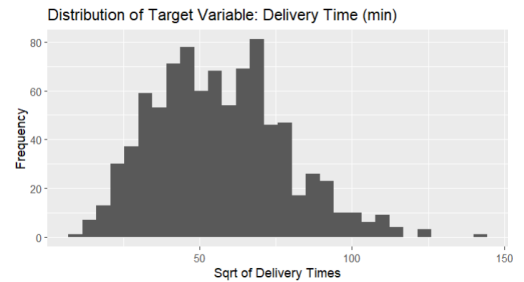


Fig. 1. Distribution of Target Variable (Delivery Time) Before Transformation.

C. Model Construction, Evaluation, and Validation

a) *Forward and Backward Selection with the Akaike Information Criterion (AIC)*: In preparation for the feature selection algorithms, the data was split into a training and test set, each holding 75 and 25 percent of the data respectively. Stratification of the data was proven to be successful with the means of each set being approximately equal (7.35 and 7.40 for the training and test set, accordingly) [3].

From there, the full model was formed, and included all possible predictors in company with the two interactions previously mentioned. With a total of 27 predictors, this

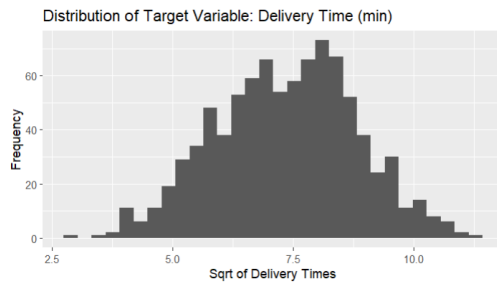


Fig. 2. Distribution of Target Variable (Delivery Time) with the Square-Root Transformation.



Fig. 3. Bivariate Data Exploration - Heat Map of Correlation Matrix.

Weather_Windy	mean	median	n
0	2.374803	2.440157	796
1	2.272918	2.342154	86
Weather_Foggy	mean	median	n
0	2.336038	2.382134	784
1	2.816077	2.881186	10
Weather_Rainy	mean	median	n
0	2.311385	2.416198	698
1	2.530180	2.549814	188
Weather_Sunny	mean	median	n
0	2.388685	2.448875	796
1	2.678883	2.748818	86
Traffic_Level_High	mean	median	n
0	2.224584	2.211103	701
1	2.362106	2.352254	181
Traffic_Level_Low	mean	median	n
0	2.543851	2.549814	136
1	2.088253	2.181828	140
Time_of_Day_Night	mean	median	n
0	2.338475	2.483375	805
1	2.205317	2.000000	76
Time_of_Day_Afternoon	mean	median	n
0	2.388685	2.448875	125
1	2.688889	2.883315	117
Vehicle_Type_Car	mean	median	n
0	2.347178	2.416198	711
1	2.431453	2.483315	173
Vehicle_Type_Scoter	mean	median	n
0	2.301015	2.483315	622
1	2.287733	2.311385	268
Customer_Experience_veryL	mean	median	n
0	2.433004	2.488889	796
1	2.688889	2.748889	86
Customer_Experience_L	mean	median	n
0	2.388889	2.488889	796
1	2.501161	2.549814	98
Customer_Experience_veryH	mean	median	n
0	2.438889	2.483315	796
1	2.588889	2.633333	76
Customer_Experience_H	mean	median	n
0	2.384051	2.416198	812
1	2.544522	2.549814	70
Customer_Experience_veryH	mean	median	n
0	2.352890	2.416198	801
1	2.684149	2.688889	85
Customer_Experience_H	mean	median	n
0	2.388889	2.483315	796
1	2.688889	2.748889	86
Customer_Experience_veryL	mean	median	n
0	2.388889	2.483315	796
1	2.688889	2.748889	86

Fig. 4. Mean and Median Values of Delivery, Split by the Levels in Each Categorical Predictor.

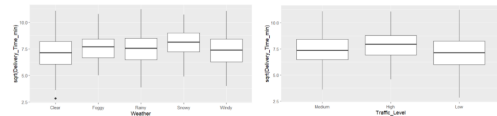


Fig. 5. Split Box Plots of Weather and Traffic Level Predictors, Respectively.

model was used in the backward selection process. Feature selection followed using a null model. Since this analysis is prediction-focused, AIC was implemented in both feature selection processes to generate a more complex model that would lead to more accurate estimates. Figure 7 reveals the performance metrics of these algorithms, showing number of predictors removed, Root Mean Squared Error (RMSE), and the adjusted R-squared values. Ultimately, the forward-selected model was the most useful out of all four model options. Since both selected models performed approximately the same in terms of how well the predicted values produced from the models align with the actual observed values in the original data set (RMSE) and what percentage of the target

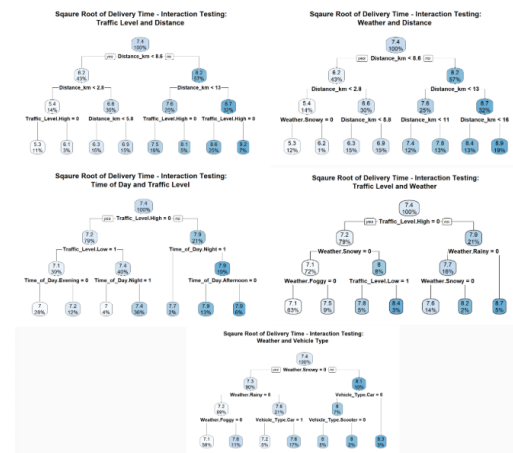


Fig. 6. Decision Trees Showing Potential Interactions.

variable's variation that the model's predictors can explain (Adj. R-squared) [6], the forward selected model was chosen due to there being less predictors, and therefore, a less complex final model which allows for more interpretability of the model while also holding accuracy in the estimates.

However, although the model was deemed most useful, it had predictors with insignificant values and the following model assumption violations: homoscedasticity, normality, and linearity. These violations are referenced in Figures 8 and 9 where the points on the Residual vs. Fitted plot have a cone-shaped trend and where the line strays from normality in the QQ Residuals plot.

Model	# Predictors Removed	RMSE	Adj. R-Squared
Null Model (benchmark)	~	1.4108	~
Full Model	~	0.6714	0.8336
Final Model (Backward Selection)	6	0.6688	0.8346
Final Model (Forward Selection)	10	0.6681	0.8340

Fig. 7. Performance Metrics Results of Backward and Forward Selection.

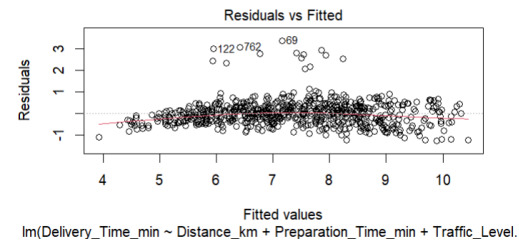


Fig. 8. Residuals vs. Fitted Plot - Forward Selected Model.

b) Ridge And Lasso Regression: Based on cross-validation, the optimal lambda values selected were 1.63 for Ridge (at alpha = 0) and 0.089 for Lasso (at alpha = 1). The higher lambda of the Ridge model indicates that it needed stronger regularization to minimize overfitting while retaining all predictors. This is consistent with Ridge's tendency to shrink the coefficients of less important variables without

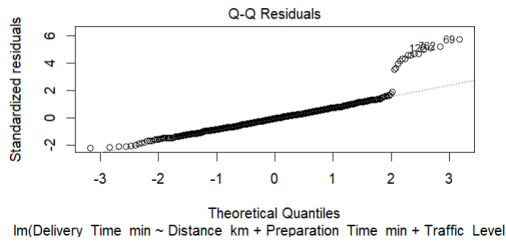


Fig. 9. QQ Residual Plot - Forward Selected Model.

setting them to zero; this is useful when many predictors contribute minimal effects.

However, Lasso performed best with a smaller lambda, applying light regularization, and suggesting that the model benefited from emphasizing only a small subset of strong predictors while minimizing or eliminating weaker ones. This also reflects Lasso's strength in producing sparse models by setting some coefficients to zero.

When comparing accuracy, the Mean Squared Error (MSE) for Lasso was 102.77, slightly lower than Ridge's 109.34. This indicates that Lasso had slightly better predictive performance in the validation set. However, the difference is not significant enough to definitively prefer one over the other based solely on MSE.

From the error plots, Ridge showed stable performance in a broader lambda range, while Lasso's performance deteriorated quickly with heavier regularization. Given the goal of reliable prediction and the potential for multicollinearity among predictors, Ridge is the more appropriate model. It maintains all the variables in the model, offering better stability and consistency - key advantages when the goal is to accurately predict delivery time using a wide range of features.

Overall, the Ridge regression model performed moderately well in predicting the final food delivery time, achieving a mean squared error (MSE) of 109.34. Its relatively high optimal lambda value indicates that strong regularization was beneficial, helping to reduce overfitting by shrinking many small coefficients without eliminating them. This suggests that multiple predictors contribute to delivery time, each with a modest effect. While Ridge did not outperform Lasso in terms of MSE, its stability and ability to handle multicollinearity make it a strong candidate for reliable prediction, especially when the goal is to retain all features in the model.

```
"Best lambda for ridge: 1.63238594113311"
"Best lambda for lasso: 0.0891654453671961"
"MSE for Ridge: 109.34441665624"
"MSE for Lasso: 102.773083665261"
```

Fig. 10. Lambda and MSE Values.

c) *Generalized Linear Model (GLM)*: Since the response variable, delivery time (Delivery_Time_min) is a continuous variable and not a discrete or binary one, logit and probit link

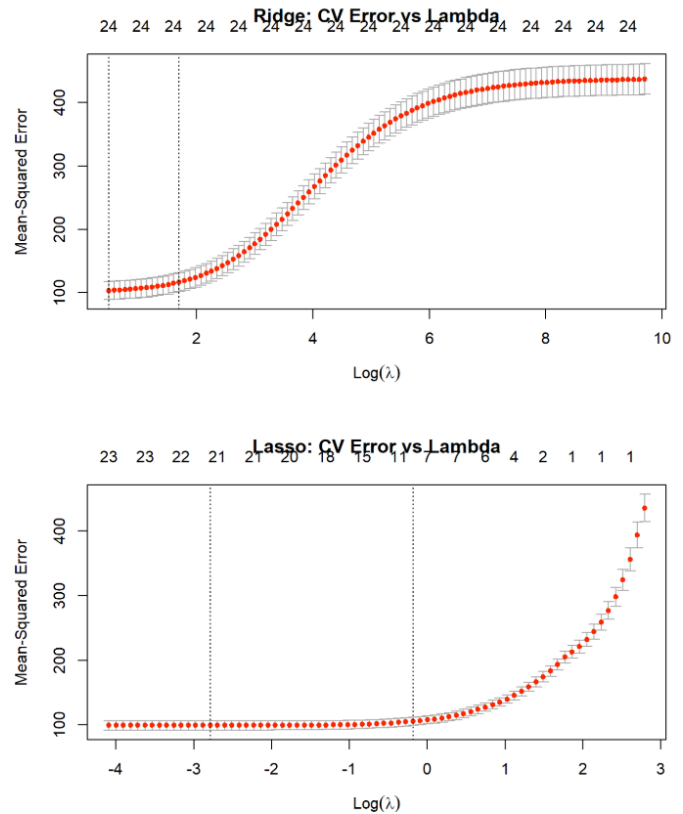


Fig. 11. Error Plots.

functions could not be implemented. Instead, the link functions chosen to experiment with were gamma and Gaussian.

For both Generalized Linear Models with gamma and Gaussian link functions, respectively, an offset was considered of distance in kilometers (Distance_km) because it has the most intuitive connection with the response variable. However, once the offset was removed and added again as $\log(\text{Distance_km})$, the AIC was worse compared to the no-offset model. Therefore, the offset was not included in this model.

For the gamma and Gaussian GLMs, a full model with all predictors was fitted. Then a stepwise model selection was conducted in which a null model was defined, and the model was reduced with AIC and BIC separately. The final models were a full model, null model, AIC reduced model, and BIC reduced model for both gamma and Gaussian GLM.

The AICs of the six models were compared, and the best model was the Gamma reduced-BIC model with an RMSE of 11.42187, R^2 of 0.775, MAE of 7.264443, AIC of 4897, and BIC of 4952. These metrics indicate a relatively good fit with decently low errors. The final model had 8 variables: Distance_km, Preparation_Time_min, Traffic_Level.Low, Weather.Foggy, Weather.Snowy, Traffic_Level.High, Weather.Rainy, and Courier_Experience_yrs.0. In the Residuals vs. Fitted Plot (Fig. 12), there is a curved pattern in the residuals, which suggests nonlinearity in the model fit. There is a slight, but not severe, violation of homoscedasticity.

In the Q-Q Residuals Plot (Fig. 13), residuals in the rightmost tail deviate from the plot. This suggests a possible violation of normality; however, the majority of the data follows the line.

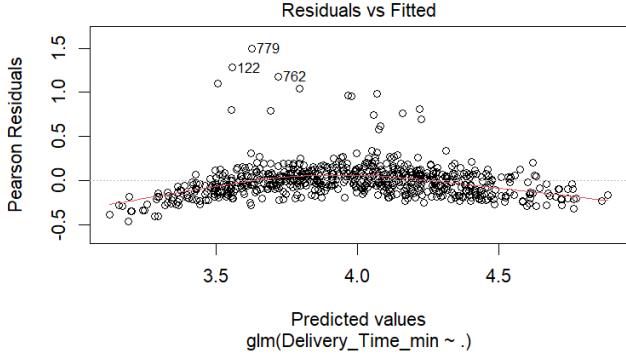


Fig. 12. Gamma GLM Residuals vs. Fitted Plot.

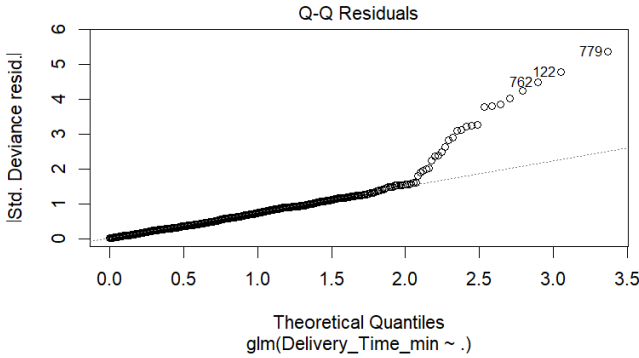


Fig. 13. Gamma GLM Q-Q Residuals Plot.

d) *Principal Component Analysis (PCA)*: PCA was applied to the full predictor set (excluding `Delivery_Time_min`) to reduce dimensionality and identify key variable influences. The data was standardized before analysis. The scree plot (Fig. 14) indicates that the first principal component (PC1) explains 6.9% of the variance, followed closely by PC2 at 6.6%. Even with the top 10 components, only 59% of the variance is captured, suggesting a diffuse structure with no dominant underlying dimension.

From the loading plots (Fig. 17 and Fig. 18), the top contributing variables to PC1 and PC2 include `Time_of_Day.Evening` (contribution $\approx 18\%$), `Time_of_Day.Afternoon` ($\approx 16\%$), `Weather.Snowy` ($\approx 14\%$), `Vehicle_Type.Car` ($\approx 13\%$), and `Traffic_Level.Low` ($\approx 11\%$). These variables had the longest vectors and highest loadings in the PCA space, indicating a stronger influence on the component directions.

The biplot (Fig. 15) shows that the data points are densely clustered around the origin, with limited visual separation along PC1 and PC2. This suggests that PCA alone may not uncover a strong latent structure in the data.

In conclusion, while PCA helped identify a few influential features, the low explained variance per component and lack

of clear grouping reinforce earlier findings: delivery time is affected by many weakly contributing factors. PCA may still be useful as a preprocessing step for regularized models or clustering to mitigate multicollinearity and reduce noise.

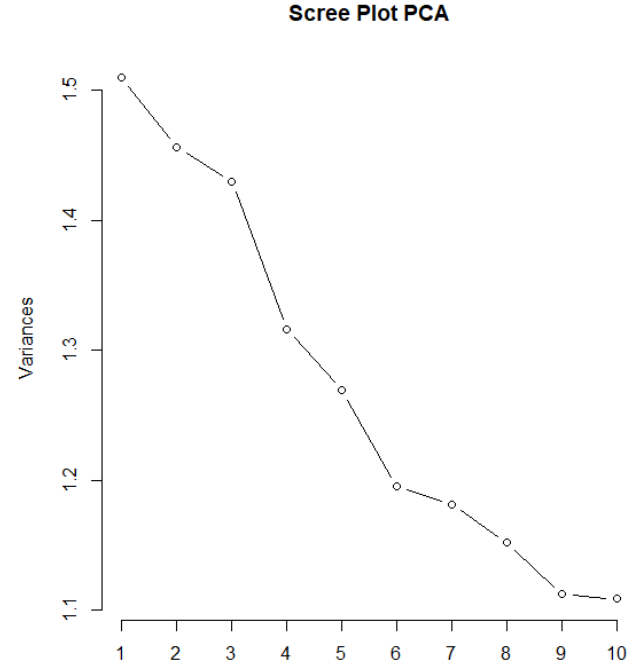


Fig. 14. PCA Scree Plot.

IV. CONCLUSION

This study aimed to understand what factors most significantly influence food delivery time. Across multiple modeling techniques, weather conditions—particularly snowy weather—and high traffic levels emerged as key predictors. For example, snowy weather and high traffic both showed differences greater than 0.5 minutes in delivery time when compared to other levels, highlighting their predictive strength. Courier experience and time of day (especially afternoon and evening) also played notable roles.

Among the models tested, the Lasso regression produced the lowest Root Mean Squared Error (RMSE) of 102.77, slightly outperforming Ridge (109.34) and the forward-selected linear model (RMSE = 103.94). The Generalized Linear Model (GLM) using a Gamma distribution also showed strong performance, with an RMSE of 112.41 and $R^2 = 0.775$.

The PCA results revealed no dominant underlying structure, with PC1 explaining only 6.9 percent of the variance and the top 10 components together accounting for just 59 percent. This suggests that delivery time is influenced by a wide range of weakly contributing variables, and no single dimension captures the majority of variation.

Overall, the project confirms that delivery time is a multi-factorial outcome best predicted by models that account for

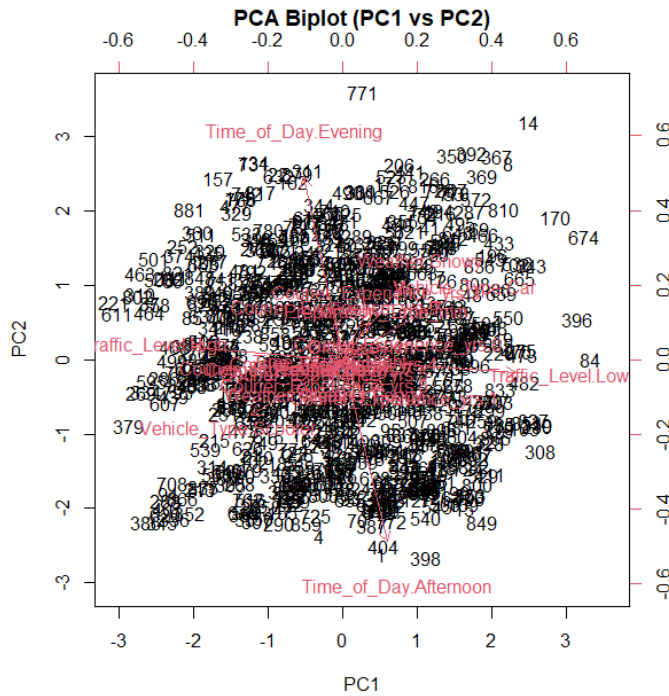


Fig. 15. PCA Biplot.

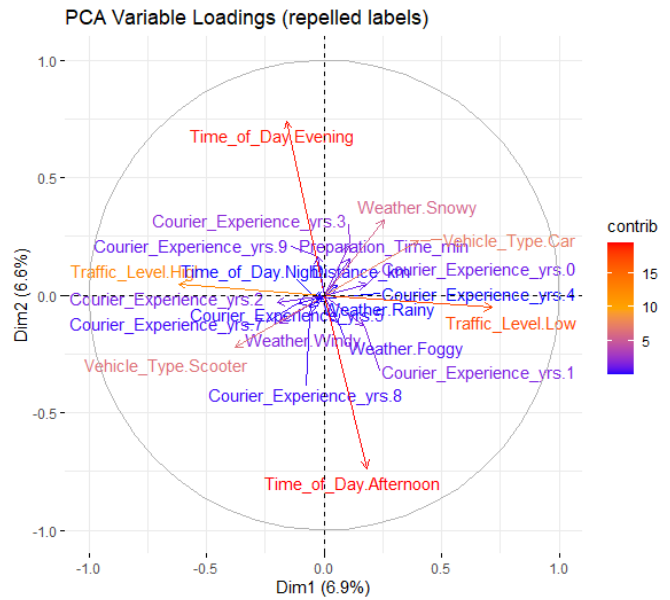


Fig. 17. PCA Variable Loadings.

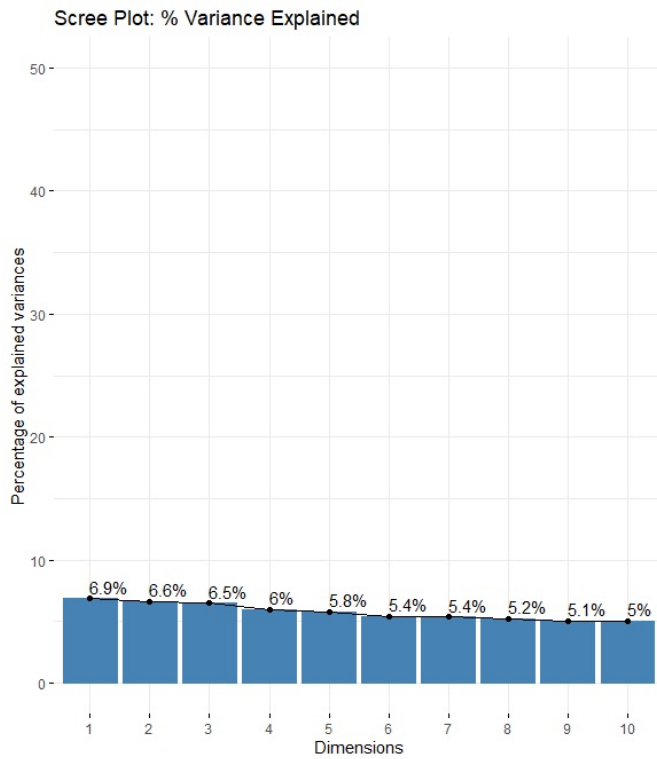


Fig. 16. Scree Plot % Variance Explained.

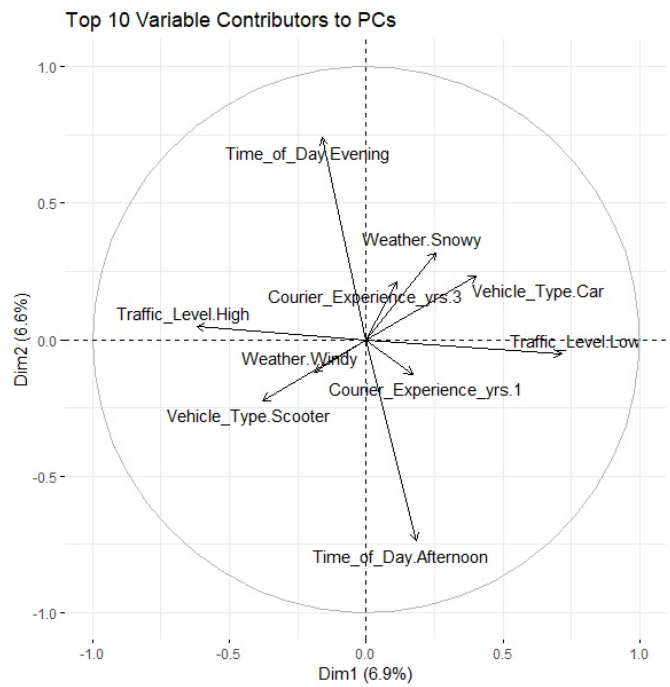


Fig. 18. PCA Variable Contributors.

complex interactions. These findings can be used to inform food delivery services in improving route planning, courier assignment, and delivery time estimates under different traffic and weather conditions.

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